

1. Problem description and structural analysis

1.1. Problem description

The Service-Level-Aware Integrated Scheduling Problem (SL-ISP) considers a finite service window containing a set of jobs $\mathcal{J} = \{1, \dots, n\}$ to be processed on a set of flexible machines $\mathcal{M} = \{1, \dots, m\}$. Each job $j \in \mathcal{J}$ belongs to a service entity (zone) identified by the mapping $g(j) \in \mathcal{R}$, where $\mathcal{R} = \{1, \dots, |\mathcal{R}|\}$ is the set of service entities. Job j is characterized by its release time $r_j \geq 0$, its quantity contribution $q_j > 0$, and a linearly ordered sequence of n_j operations $\mathcal{O}_j = \{O_{j1}, O_{j2}, \dots, O_{jn_j}\}$. Operation O_{jo} can be processed on any machine h from a job- and operation-specific eligible set $\mathcal{M}_{jo} \subseteq \mathcal{M}$, with machine-dependent processing time $p_{joh} > 0$. Operations within each job obey strict linear precedence; operations belonging to different jobs may be processed concurrently subject to machine availability. Preemption is not permitted.

Each service entity $r \in \mathcal{R}$ carries a service-level commitment defined by four parameters: a common delivery deadline d_r , a deterministic transportation delay $\tau_r \geq 0$, a minimum required fulfillment ratio $\rho_r \in (0, 1]$, and a relative importance weight $w_r > 0$. The aggregate committed quantity for entity r is

$$Q_r = \sum_{j: g(j)=r} q_j, \quad (1)$$

and the minimum required on-time quantity is

$$Q_r^{\min} = \rho_r Q_r. \quad (2)$$

Let C_j denote the production completion time of the final operation of job j . Incorporating the entity-specific transport delay, the delivery time is

$$D_j = C_j + \tau_{g(j)}. \quad (3)$$

Job j is considered on time if $D_j \leq d_{g(j)}$; otherwise it incurs tardiness

$$T_j = [D_j - d_{g(j)}]_+, \quad (4)$$

where $[x]_+ = \max(0, x)$. The total on-time fulfilled quantity of entity r is

$$Q_r^{\text{on}} = \sum_{j: g(j)=r, D_j \leq d_r} q_j, \quad (5)$$

and the service shortfall is

$$U_r = [Q_r^{\min} - Q_r^{\text{on}}]_+. \quad (6)$$

The system operates under event-driven rescheduling. At each decision epoch—triggered by a job release, an operation completion, or a machine becoming idle—the scheduler assigns a waiting operation to an available machine or deliberately leaves the machine idle. The objective integrates operational efficiency and strategic service-level attainment:

$$\min Z = \alpha \sum_{j \in \mathcal{J}} T_j + \beta \sum_{r \in \mathcal{R}} w_r U_r, \quad (7)$$

where $\alpha, \beta > 0$ are weighting coefficients that encode the relative cost of unit tardiness versus unit weighted shortfall. The coefficient $\gamma = \beta/\alpha$ captures the service-pressure ratio: larger γ places greater emphasis on meeting entity-level service commitments.

1.2. Complexity of zero-shortfall service fulfillment

A central structural question is whether the scheduler can achieve zero shortfall ($U_r = 0$ for all r) for a given instance. Even in a maximally restricted setting, this decision problem is computationally intractable.

Definition 1 (Restricted Zero-Shortfall Service Fulfillment Problem). An instance of the Restricted Zero-Shortfall Service Fulfillment Problem (RZSSFP) is specified by: one service entity, a single machine, a set of single-operation jobs $\mathcal{J}$ all released at time zero, zero transportation delay, and a common deadline D . Each job $j \in \mathcal{J}$ has processing time p_j and quantity q_j . The required on-time quantity is Q . The question is: does there exist a subset of jobs whose total processing time does not exceed D and whose total quantity is at least Q ?

Theorem 1. *RZSSFP is NP-complete.*

PROOF. Membership in NP. A certificate consists of a subset $S \subseteq \mathcal{J}$. Verification requires checking $\sum_{j \in S} p_j \leq D$ and $\sum_{j \in S} q_j \geq Q$, both computable in $O(|\mathcal{J}|)$ time.

NP-hardness. We reduce from the 0-1 Knapsack Decision Problem (KDP), which is well known to be NP-complete (Garey and Johnson, 1979). An instance of KDP consists of n items, each with weight w_i and value v_i , a knapsack capacity W , and a target value V . The question is whether there exists a subset $S \subseteq \{1, \dots, n\}$ such that $\sum_{i \in S} w_i \leq W$ and $\sum_{i \in S} v_i \geq V$.

Construct an RZSSFP instance as follows. For each item i in the KDP instance, create a job j with $p_j = w_i$, $q_j = v_i$, and $r_j = 0$. Set the common deadline $D = W$, the required on-time quantity $Q = V$, and the transport delay $\tau = 0$. In this single-machine setting with all jobs available at time zero, a set of jobs completes by D if and only if the sum of their processing times does not exceed D . Hence the KDP instance admits a feasible subset if and only if the constructed RZSSFP instance admits zero shortfall. The reduction requires $O(n)$ time.

Since RZSSFP is in NP and is NP-hard, it is NP-complete.

Theorem 1 establishes that even the simplest special case of zero-shortfall verification is computationally intractable. This justifies the use of heuristic and metaheuristic methods for the full SL-ISP, and motivates the development of computationally efficient surrogate indicators—rather than exact feasibility tests—for guiding online scheduling decisions.

1.3. Optimistic service-recovery relaxation

To enable tractable online diagnosis of service recoverability, we introduce an optimistic relaxation that ignores future job releases, inter-machine interference, and sequencing constraints beyond a designated bottleneck resource set $\mathcal{B} \subseteq \mathcal{M}$. The set $\mathcal{B}$ models the *unavoidable resource pool*: machines whose capacity is structurally constrained and through which recovery-oriented jobs must pass. Typical choices for $\mathcal{B}$ include the set of machines with the highest average utilization, or machines that are uniquely eligible for a large fraction of operations.

At any decision time t during online execution, we define the following state-dependent quantities:

- $Q_r^{sec}(t)$: **secured on-time quantity** of entity r —the total quantity of entity- r jobs already delivered with $D_j \leq d_r$ by time t .
- $Q_r^{rem}(t) = [Q_r^{min} - Q_r^{sec}(t)]_+$: **remaining required quantity** for entity r .
- $\underline{D}_j(t)$: **optimistic delivery time lower bound** for job j . For a job whose remaining operations at time t are $\mathcal{O}_j^{rem}(t)$, we define

$$\underline{D}_j(t) = t + \sum_{o \in \mathcal{O}_j^{rem}(t)} \min_{h \in \mathcal{M}_{j_o}} p_{joh} + \tau_{g(j)}, \quad (8)$$

where the processing time of a partially completed operation is replaced by its residual. This bound assumes zero waiting time and the fastest eligible machine for every operation.

- $\mathcal{E}_r(t) = \{j \in \mathcal{J} : g(j) = r, \underline{D}_j(t) \leq d_r\}$: the **eligible set**—jobs of entity r that can optimistically meet the deadline d_r .
- $\text{Cap}_{\mathcal{B}}(t, d_r) = \sum_{h \in \mathcal{B}} \text{Cap}_h(t, d_r)$: **aggregate available bottleneck capacity** during the interval $[t, d_r]$, where $\text{Cap}_h(t, d_r)$ is the remaining processing time on machine h net of already-committed workload within $[t, d_r]$.

- $\underline{w}_{j,B}(t)$: **minimum bottleneck workload** of job j , defined as

$$\underline{w}_{j,B}(t) = \sum_{o \in \mathcal{O}_j^{rem}(t)} \min_{h \in B \cap \mathcal{M}_{j_o}} p_{joh}, \quad (9)$$

with the convention that $\min_{h \in \emptyset} p_{joh} = 0$. This is the smallest total processing time that job j must consume on machines in B to complete its remaining operations.

These definitions support two complementary knapsack-type optimization problems. The **minimum-workload recovery problem** seeks the smallest bottleneck workload required to assemble enough on-time quantity:

$$W_{r,B}^{min}(t) = \min \sum_{j \in \mathcal{E}_r(t)} \underline{w}_{j,B}(t) y_j \quad (10)$$

subject to

$$\sum_{j \in \mathcal{E}_r(t)} q_j y_j \geq Q_r^{rem}(t), \quad y_j \in \{0, 1\}. \quad (11)$$

If no subset of $\mathcal{E}_r(t)$ can meet the quantity requirement, we set $W_{r,B}^{min}(t) = +\infty$.

The **maximum recoverable service quantity problem** maximizes the on-time deliverable quantity subject to the bottleneck capacity constraint:

$$\overline{Q}_{r,B}^{rec}(t) = \max \sum_{j \in \mathcal{E}_r(t)} q_j z_j \quad (12)$$

subject to

$$\sum_{j \in \mathcal{E}_r(t)} \underline{w}_{j,B}(t) z_j \leq \text{Cap}_B(t, d_r), \quad z_j \in \{0, 1\}. \quad (13)$$

Problems (10) and (12) form a dual pair of 0-1 knapsack problems: the former minimizes workload for a given quantity target, while the latter maximizes quantity for a given workload budget. Both can be solved in pseudo-polynomial time via standard dynamic programming, making them suitable for repeated online evaluation at each rescheduling epoch.

1.4. Dual knapsack characterizations

The following theorem establishes the formal equivalence between the two optimistic recoverability indicators.

Theorem 2 (Equivalent characterizations of optimistic zero-shortfall recoverability). *For any entity $r \in \mathcal{R}$ at any decision time t ,*

$$W_{r,B}^{min}(t) \leq \text{Cap}_B(t, d_r) \iff \overline{Q}_{r,B}^{rec}(t) \geq Q_r^{rem}(t). \quad (14)$$

PROOF. ($\Rightarrow$) Assume $W_{r,B}^{min}(t) \leq \text{Cap}_B(t, d_r)$. Let y^* be an optimal solution to problem (10). Then

$$\sum_{j \in \mathcal{E}_r(t)} \underline{w}_{j,B}(t) y_j^* = W_{r,B}^{min}(t) \leq \text{Cap}_B(t, d_r)$$

and $\sum_{j \in \mathcal{E}_r(t)} q_j y_j^* \geq Q_r^{rem}(t)$. Defining $z_j = y_j^*$ for all $j \in \mathcal{E}_r(t)$, the vector z satisfies the capacity constraint of problem (12) and yields objective value at least $Q_r^{rem}(t)$. Since $\overline{Q}_{r,B}^{rec}(t)$ is the maximum over all feasible solutions,

$$\overline{Q}_{r,B}^{rec}(t) \geq \sum_j q_j z_j \geq Q_r^{rem}(t).$$

($\Leftarrow$) Assume $\overline{Q}_{r,B}^{rec}(t) \geq Q_r^{rem}(t)$. Let z^* be an optimal solution to problem (12). Then

$$\sum_{j \in \mathcal{E}_r(t)} q_j z_j^* = \overline{Q}_{r,B}^{rec}(t) \geq Q_r^{rem}(t)$$

and $\sum_{j \in \mathcal{E}_r(t)} w_{j,B}(t) z_j^* \leq \text{Cap}_B(t, d_r)$. Defining $y_j = z_j^*$ for all $j \in \mathcal{E}_r(t)$, the vector y satisfies the quantity constraint of problem (10) and is therefore a feasible—but not necessarily optimal—solution. Hence

$$W_{r,B}^{min}(t) \leq \sum_j w_{j,B}(t) z_j^* \leq \text{Cap}_B(t, d_r).$$

Both directions hold, establishing the equivalence.

Theorem 2 provides two computationally complementary routes for evaluating optimistic recoverability. One may either check whether the minimum required workload fits within the remaining bottleneck capacity (workload view), or whether the maximum recoverable quantity meets the remaining requirement (quantity view). Both characterizations are exact with respect to the optimistic relaxation and yield identical yes/no answers. In practice, the quantity-maximization formulation (12) is often more convenient, as $\text{Cap}_B(t, d_r)$ is directly observable from the schedule state.

1.5. Unavoidable service shortfall lower bound

While the optimistic relaxation neglects sequencing interference and machine contention, the quantity $\overline{Q}_{r,B}^{rec}(t)$ nonetheless yields a valid lower bound on the shortfall achievable by any feasible continuation of the current schedule.

Theorem 3 (Unavoidable shortfall lower bound). *For any feasible continuation schedule π from time t , the service shortfall of entity r satisfies*

$$U_r^\pi \geq \underline{U}_{r,B}(t) = [Q_r^{rem}(t) - \overline{Q}_{r,B}^{rec}(t)]_+. \quad (15)$$

PROOF. Let $J_r^{on}(\pi) \subseteq \mathcal{E}_r(t)$ denote the set of entity- r jobs that are not yet delivered at time t but are delivered on time ($D_j^\pi \leq d_r$) under continuation π . Every such job must complete its remaining operations, and the total workload assigned to machines in B cannot exceed the available capacity during $[t, d_r]$. Therefore

$$\sum_{j \in J_r^{on}(\pi)} w_{j,B}(t) \leq \text{Cap}_B(t, d_r),$$

which means the indicator vector of $J_r^{on}(\pi)$ is a feasible solution to the maximum recoverable service quantity problem (12). Consequently,

$$\sum_{j \in J_r^{on}(\pi)} q_j \leq \overline{Q}_{r,B}^{rec}(t).$$

The total on-time quantity delivered under π is $Q_r^{sec}(t) + \sum_{j \in J_r^{on}(\pi)} q_j$. The resulting shortfall is

$$\begin{aligned} U_r^\pi &= [Q_r^{min} - Q_r^{sec}(t) - \sum_{j \in J_r^{on}(\pi)} q_j]_+ \\ &= [Q_r^{rem}(t) - \sum_{j \in J_r^{on}(\pi)} q_j]_+ \\ &\geq [Q_r^{rem}(t) - \overline{Q}_{r,B}^{rec}(t)]_+ = \underline{U}_{r,B}(t), \end{aligned}$$

where the inequality follows from $\sum_{j \in J_r^{on}(\pi)} q_j \leq \overline{Q}_{r,B}^{rec}(t)$ and the monotonicity of $[\cdot]_+$.

Theorem 3 is structural rather than algorithmic: it does not require solving the full scheduling problem to evaluate the bound. Instead, the lower bound $\underline{U}_{r,B}(t)$ is obtained by solving a single 0-1 knapsack problem per entity. This yields a computationally lightweight certificate that is valid for *all* feasible continuations.

Corollary 1 (Unattainability of zero shortfall). *If $\overline{Q}_{r,B}^{rec}(t) < Q_r^{rem}(t)$ at time t , then $\underline{U}_{r,B}(t) > 0$, and by Theorem 3, $\underline{U}_r^\pi > 0$ for every feasible continuation schedule π . Thus, zero shortfall for entity r is provably unattainable regardless of future scheduling decisions.*

Corollary 1 provides an early and computable certificate of unavoidable service failure. When triggered, it enables the scheduler to reallocate computational and physical resources toward entities for which recovery remains possible, rather than pursuing an impossible target.

1.6. Recoverability erosion

A natural question for online control is how the optimistic recoverability indicators evolve as time advances and scheduling decisions are executed. Intuitively, the consumption of bottleneck capacity by operations that do not contribute to a given entity's recovery should degrade that entity's recoverability. The following theorem formalizes this intuition under mild and practically typical conditions.

Theorem 4 (Recoverability erosion under non-recovery capacity consumption). *Let $t_1 < t_2$ be two decision epochs. Assume the following conditions hold for entity r :*

- (C1) $Q_r^{rem}(t_2) = Q_r^{rem}(t_1)$ (remaining required quantity is unchanged);
- (C2) $\mathcal{E}_r(t_2) \subseteq \mathcal{E}_r(t_1)$ (the eligible set does not expand);
- (C3) $\underline{w}_{j,B}(t_2) \geq \underline{w}_{j,B}(t_1)$ for all $j \in \mathcal{E}_r(t_2)$ (the minimum bottleneck workload per job is non-decreasing);
- (C4) $\text{Cap}_B(t_2, d_r) \leq \text{Cap}_B(t_1, d_r)$ (available bottleneck capacity is non-increasing).

Then the following monotonicity properties hold:

$$\overline{Q}_{r,B}^{rec}(t_2) \leq \overline{Q}_{r,B}^{rec}(t_1), \quad (16)$$

$$\underline{U}_{r,B}(t_2) \geq \underline{U}_{r,B}(t_1). \quad (17)$$

PROOF. Let z^* be an optimal solution to the maximum recoverable service quantity problem at time t_2 , so that $\overline{Q}_{r,B}^{rec}(t_2) = \sum_{j \in \mathcal{E}_r(t_2)} q_j z_j^*$. By condition (C2), every job selected by z^* also belongs to $\mathcal{E}_r(t_1)$. Construct a candidate solution z' for time t_1 by setting $z'_j = z_j^*$ for $j \in \mathcal{E}_r(t_2)$ and $z'_j = 0$ for $j \in \mathcal{E}_r(t_1) \setminus \mathcal{E}_r(t_2)$. Its workload consumption at t_1 satisfies

$$\begin{aligned} \sum_{j \in \mathcal{E}_r(t_1)} \underline{w}_{j,B}(t_1) z'_j &= \sum_{j \in \mathcal{E}_r(t_2)} \underline{w}_{j,B}(t_1) z_j^* \\ &\leq \sum_{j \in \mathcal{E}_r(t_2)} \underline{w}_{j,B}(t_2) z_j^* \quad (\text{by C3}) \\ &\leq \text{Cap}_B(t_2, d_r) \quad (\text{feasibility of } z^* \text{ at } t_2) \\ &\leq \text{Cap}_B(t_1, d_r) \quad (\text{by C4}). \end{aligned}$$

Thus z' is feasible for the problem at t_1 , yielding

$$\overline{Q}_{r,B}^{rec}(t_1) \geq \sum_{j \in \mathcal{E}_r(t_1)} q_j z'_j = \sum_{j \in \mathcal{E}_r(t_2)} q_j z_j^* = \overline{Q}_{r,B}^{rec}(t_2),$$

which proves (16).

For (17), applying the definition of the lower bound together with condition (C1) and inequality (16):

$$\begin{aligned} \underline{U}_{r,B}(t_2) &= [Q_r^{rem}(t_2) - \overline{Q}_{r,B}^{rec}(t_2)]_+ \\ &\geq [Q_r^{rem}(t_1) - \overline{Q}_{r,B}^{rec}(t_1)]_+ = \underline{U}_{r,B}(t_1). \end{aligned}$$

Theorem 4 captures the fundamental directional pressure in online service-level management: under quiescent conditions (no new job arrivals for entity r , no capacity replenishment), the passage of time degrades recoverability. Conditions (C1)–(C4) describe the typical regime between job-release events for a given entity: no additional quantity

has been secured, the eligible set shrinks as remaining slack diminishes, residual workloads do not decrease, and bottleneck capacity is progressively consumed. The result underscores the time-critical nature of recovery-oriented intervention: delaying recovery action can only weaken the achievable outcome, never improve it.

It is important to note that Theorem 4 does *not* claim that recoverability always decreases monotonically over time. When new jobs arrive (violating C2) or capacity is freed by completion of non-bottleneck work, recoverability may increase. The conditions (C1)–(C4) delineate precisely when the monotonic decrease is guaranteed.

1.7. Mandatory rescue structure

When optimistic zero-shortfall recovery is still attainable for entity r , the recovery may be *fragile*: certain jobs are indispensable, and their exclusion would render the quantity target infeasible. Identifying these jobs is essential for constructing effective recovery actions.

For a candidate job $j \in \mathcal{E}_r(t)$, define the **exclusion-based maximum recoverable quantity**:

$$\overline{Q}_{r,B}^{rec,-j}(t) = \max \sum_{i \in \mathcal{E}_r(t) \setminus \{j\}} q_i z_i \quad (18)$$

subject to

$$\sum_{i \in \mathcal{E}_r(t) \setminus \{j\}} \underline{w}_{i,B}(t) z_i \leq \text{Cap}_B(t, d_r), \quad z_i \in \{0, 1\}. \quad (19)$$

This is the maximum on-time quantity achievable when job j is excluded from consideration.

Definition 2 (Mandatory rescue job). A job $j \in \mathcal{E}_r(t)$ is a **mandatory rescue job** for entity r at time t if

$$\overline{Q}_{r,B}^{rec}(t) \geq Q_r^{rem}(t) \quad \text{but} \quad \overline{Q}_{r,B}^{rec,-j}(t) < Q_r^{rem}(t).$$

In words, optimistic zero-shortfall recovery is feasible when j is available, but becomes infeasible the moment j is excluded. Mandatory rescue jobs are the load-bearing elements of the recovery plan.

Theorem 5 (Mandatory on-time delivery). *If j is a mandatory rescue job for entity r at time t , then every feasible continuation schedule π with $U_r^\pi = 0$ must deliver j no later than d_r .*

PROOF. Suppose, for contradiction, that there exists a feasible continuation schedule π achieving $U_r^\pi = 0$ in which job j is delivered after the deadline ($D_r^\pi > d_r$). Let $\mathcal{J}_r^{on}(\pi) \subseteq \mathcal{E}_r(t)$ be the set of entity- r jobs delivered on time under π . Since j is late, $\mathcal{J}_r^{on}(\pi) \subseteq \mathcal{E}_r(t) \setminus \{j\}$.

Because $U_r^\pi = 0$, the on-time quantity from these jobs must satisfy the remaining requirement:

$$\sum_{i \in \mathcal{J}_r^{on}(\pi)} q_i \geq Q_r^{rem}(t).$$

As in the proof of Theorem 3, the capacity feasibility of π implies that the indicator vector of $\mathcal{J}_r^{on}(\pi)$ satisfies the capacity constraint of problem (18). Hence it is a feasible solution to the exclusion-based problem, yielding

$$\overline{Q}_{r,B}^{rec,-j}(t) \geq \sum_{i \in \mathcal{J}_r^{on}(\pi)} q_i \geq Q_r^{rem}(t),$$

which contradicts the definition of a mandatory rescue job ($\overline{Q}_{r,B}^{rec,-j}(t) < Q_r^{rem}(t)$). Therefore, no such continuation π can exist: any zero-shortfall schedule must deliver j on time.

Theorem 5 provides a rigorous justification for elevating mandatory rescue jobs to the highest priority in recovery-oriented dispatching. We now develop two practically verifiable sufficient conditions and an exact marginal characterization.

Proposition 1 (Quantity-mandatory jobs). Assume $\overline{Q}_{r,B}^{rec}(t) \geq Q_r^{rem}(t)$. If

$$\sum_{i \in \mathcal{E}_r(t) \setminus \{j\}} q_i < Q_r^{rem}(t), \quad (20)$$

then j is a mandatory rescue job.

PROOF. Condition (20) states that the total quantity of all eligible jobs other than j is insufficient to meet the remaining requirement. Since any feasible solution to the exclusion-based problem (18) selects only from $\mathcal{E}_r(t) \setminus \{j\}$, its objective value cannot exceed the total available quantity:

$$\overline{Q}_{r,B}^{rec,-j}(t) \leq \sum_{i \in \mathcal{E}_r(t) \setminus \{j\}} q_i < Q_r^{rem}(t).$$

Together with the premise $\overline{Q}_{r,B}^{rec}(t) \geq Q_r^{rem}(t)$, the conditions of Definition 2 are satisfied.

Quantity-mandatory jobs are indispensable for a simple reason: even if *all* other eligible jobs were delivered on time (ignoring capacity), the quantity target could not be met. Their mandatory status is driven purely by the quantity balance, independent of the capacity constraint.

Proposition 2 (Capacity-mandatory jobs). Assume $\overline{Q}_{r,B}^{rec}(t) \geq Q_r^{rem}(t)$. If

$$\sum_{i \in \mathcal{E}_r(t) \setminus \{j\}} q_i \geq Q_r^{rem}(t) \quad \text{but} \quad \overline{Q}_{r,B}^{rec,-j}(t) < Q_r^{rem}(t), \quad (21)$$

then j is a mandatory rescue job.

PROOF. The conditions in (21) directly match Definition 2 (the first inequality ensures j is not quantity-mandatory, while the second establishes the exclusion-based infeasibility).

In contrast to quantity-mandatory jobs, capacity-mandatory jobs are indispensable not because of insufficient aggregate quantity, but because the bottleneck capacity is inadequate to process any alternative combination of jobs that meets the quantity target. Their mandatory status arises from the interaction between the quantity requirement and the capacity constraint.

Proposition 3 (Marginal-recovery characterization). Define the *marginal recoverability contribution* of job j and the *recovery surplus* of entity r as

$$\Delta_{j,r,B}^{rec}(t) = \overline{Q}_{r,B}^{rec}(t) - \overline{Q}_{r,B}^{rec,-j}(t), \quad (22)$$

$$S_{r,B}^{rec}(t) = \overline{Q}_{r,B}^{rec}(t) - Q_r^{rem}(t). \quad (23)$$

Assume $\overline{Q}_{r,B}^{rec}(t) \geq Q_r^{rem}(t)$. Then j is a mandatory rescue job if and only if

$$\Delta_{j,r,B}^{rec}(t) > S_{r,B}^{rec}(t). \quad (24)$$

PROOF. By Definition 2, j is mandatory if and only if $\overline{Q}_{r,B}^{rec,-j}(t) < Q_r^{rem}(t)$. Substituting $\overline{Q}_{r,B}^{rec,-j}(t) = \overline{Q}_{r,B}^{rec}(t) - \Delta_{j,r,B}^{rec}(t)$ from (22), this inequality becomes

$$\overline{Q}_{r,B}^{rec}(t) - \Delta_{j,r,B}^{rec}(t) < Q_r^{rem}(t) \iff \Delta_{j,r,B}^{rec}(t) > \overline{Q}_{r,B}^{rec}(t) - Q_r^{rem}(t) = S_{r,B}^{rec}(t),$$

where the last equality uses (23).

Proposition 3 provides an intuitive and computationally useful characterization: a job is mandatory precisely when its removal would eliminate more recoverable quantity than the current surplus above the requirement. When the recovery margin $S_{r,B}^{rec}(t)$ is thin, even jobs with modest marginal contributions become mandatory. Conversely, when the surplus is large, the system can afford to lose individual jobs without compromising recoverability.

Implications for algorithm design. The structural results established in this section collectively motivate the core mechanisms of the proposed Recoverability-Guided Rolling-Horizon Optimization with Large Neighborhood Search (RG-RHO-LNS) algorithm:

1. The NP-completeness of zero-shortfall fulfillment (Theorem 1) justifies the use of metaheuristic search rather than exact optimization for each rolling-horizon subproblem.
2. The dual knapsack characterizations (Theorem 2) enable efficient online recoverability diagnosis: both $W_{r,B}^{min}(t)$ and $\overline{Q}_{r,B}^{rec}(t)$ are computable in pseudo-polynomial time via standard dynamic programming for 0-1 knapsack, supporting real-time entity-level assessment at every rescheduling epoch.
3. The unavoidable shortfall lower bound (Theorem 3) and its corollary identify entities for which zero shortfall is provably unattainable, enabling the algorithm to classify entities and direct computational effort toward those where recovery remains possible.
4. The recoverability erosion theorem (Theorem 4) highlights the time-critical nature of recovery actions, motivating early and aggressive intervention for entities approaching the unattainability threshold.
5. The mandatory rescue structure (Theorem 5, Propositions 1–3) directly informs the construction of the *recoverability-guided active set*: mandatory rescue jobs are included and prioritized in the repair neighborhood. Entity classification based on the recovery surplus $S_{r,B}^{rec}(t)$ —distinguishing secured, stably-recoverable, fragile-recoverable, and partially-unrecoverable entities—determines the composition of recovery-oriented neighborhoods that target specific bottleneck resources.

Together, these theoretical elements form the recoverability-guidance module of RG-RHO-LNS, establishing a principled bridge between the structural properties of SL-ISP and its online heuristic solution.